RACE
Regular Analysis Through Continuous Exercise

Limits: RACE-01

Single Correct Type Question

1. $\lim\limits_{x\to 2}\dfrac{x^3-6x^2+11x-6}{x^2-6x+8}$ is equal to

(A) $\dfrac{3}{2}$ (B) $\dfrac{1}{2}$

(C) 1 (D) Does not exist

2. $\lim\limits_{x\to 3}\dfrac{x^3-7x^2+15x-9}{x^4-5x^3+27x-27}$ is equal to

(A) 2/9 (B) 1/9

(C) –2/9 (D) Does not exist

3. $\lim\limits_{x\to 1}\left(\dfrac{1}{1-x}-\dfrac{3}{1-x^3}\right)$ is equal to

(A) –1 (B) 0

(C) 1 (D) Does not exist

4. $\underset{x\to 1}{\text{Lim}}\dfrac{\left(\sum\limits_{k=1}^{100}x^k\right)-100}{x-1}=$_____

(A) 5000 (B) 5050

(C) 5005 (D) 10100

5. $\lim\limits_{x\to 2}\dfrac{\sqrt{1+\sqrt{2+x}}-\sqrt{3}}{x-2}$ is equal to

(A) $\dfrac{1}{\sqrt{3}}$ (B) $\sqrt{3}$

(C) $\dfrac{1}{4\sqrt{3}}$ (D) $\dfrac{1}{8\sqrt{3}}$

6. $\lim\limits_{n\to\infty}(n-\sqrt{n^2+n})$ is equal to

(A) 1 (B) $\dfrac{1}{2}$

(C) $-\dfrac{1}{2}$ (D) does not exist

7. The value of $\lim\limits_{n\to\infty}\dfrac{\lfloor 3(n+1)}{(n+1)^3\lfloor 3n},n\in N$ is _____

(A) 27 (B) 9

(C) 3 (D) 81

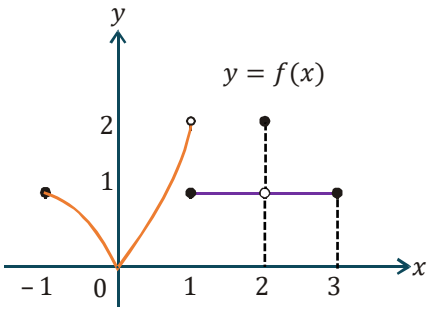
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Comprehension

Paragraph for Questions 8 and 9

Graph of $y = f(x)$ is given below. Answer the two questions based on it.



8. Which of the following are correct?

(A) $\lim\limits_{x\to1^+} f(x)=1$ (B) $\lim\limits_{x\to2} f(x)=2$

(C) $\lim\limits_{x\to1^-} f(x)=2$ (D) $\lim\limits_{x\to0^+} f(x)=\lim\limits_{x\to0^-} f(x)$

9. Which of the following are incorrect?

(A) $\lim\limits_{x\to c} f(x)$ exists at every c ∈ (–1, 1) (B) $\lim\limits_{x\to c} f(x)$ exists at every c ∈ (1, 3)

(C) $\lim\limits_{x\to1^-} f(x)=0$ (D) $\lim\limits_{x\to3^+} f(x)$ exists

Subjective Type Questions:

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10. Evaluate: $\lim\limits_{x\to a}\dfrac{\sqrt{a+2x}-\sqrt{3x}}{\sqrt{3a+x}-2\sqrt{x}}$, $a\neq0$

11. If $G(x) = -\sqrt{25-x^2}$, then find the $\lim\limits_{x\to1}\left(\dfrac{G(x)-G(1)}{x-1}\right)$

12. Find $\lim\limits_{n\to\infty}\left(1-\dfrac{1}{2^2}\right)\left(1-\dfrac{1}{3^2}\right)\left(1-\dfrac{1}{4^2}\right)...\left(1-\dfrac{1}{n^2}\right)$

13. Evaluate: $\lim\limits_{n\to\infty}\dfrac{1.n+2.(n-1)+3(n-2)...+n.1}{n^3}$

Answer Key

Question	1	2	3	4	5	6	7	8	9
Answer	B	A	A	B	D	C	A	A,C,D	C,D

10. $\dfrac{2}{3\sqrt{3}}$ **11.** $\dfrac{1}{\sqrt{24}}$ **12.** $\dfrac{1}{2}$ **13.** $\dfrac{1}{6}$

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RACE
Regular Analysis Through Continuous Exercise

Limits: RACE-02

Single Correct Type Question

1. If the value of $\lim\limits_{n\to\infty} n^3\left(1-\frac{3}{7}\right)\left(1-\frac{3}{8}\right)\left(1-\frac{3}{9}\right)...\left(1-\frac{3}{n}\right)$ is k, then sum of the digits of k is?

(A) 6 (B) 2 (C) 3 (D) 8

2. $\lim\limits_{x\to 0}\frac{(1-\cos 2x)\sin 5x}{x^2\sin 3x}=$

(A) 10/3 (B) 3/10 (C) 6/5 (D) 5/6

3. $\lim\limits_{x\to 1^-}\frac{1-\sqrt{x}}{\left(\cos^{-1}x\right)^2}$ is equal to

(A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{4}$

4. $\lim\limits_{x\to 0}\frac{64}{x^8}\left[1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right]=$_____

(A) 0.25 (B) 0.50 (C) 0.75 (D) 1

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5. $\lim\limits_{x\to 0}\frac{e^{x^2}-\cos x}{x^2}=$_____

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $e-1$

Matrix Match Type Questions

6.

Column-I		Column-II	
(P)	$\lim\limits_{x\to 0}\frac{2^x-1}{(1+x)^{1/2}-1}$	(1)	2
(Q)	$\lim\limits_{x\to 0}\frac{e^{3x}-1}{x/2}$	(2)	2 ln2
(R)	$\lim\limits_{x\to 0}\frac{e^{\sin 2x}-e^{\sin x}}{x}$	(3)	6
(S)	$\lim\limits_{x\to 0}\frac{e^x+e^{-x}-2}{x^2}$	(4)	1

(A) P→3; Q → 4; R → 2; S→1
(B) P→2; Q → 4; R → 1; S→2
(C) P→3; Q → 2; R → 2; S→4
(D) P→2; Q → 3; R → 4; S→4

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Integer Type Question

7. The value of $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{1-\tan x}{1-\sqrt{2}\sin x}$ is?

8. $\lim\limits_{n\to\infty}\dfrac{(n+1)^4-(n-1)^4}{(n+1)^4+(n-1)^4}$ is equal to

9. Using Sandwich theorem, evaluate

$$\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+.....+\frac{1}{\sqrt{n^2+2n}}\right)$$

Subjective Type Question

10. Evaluate $\lim\limits_{x\to0}\left[\dfrac{-2x}{\tan x}\right]$, where [.] denotes greatest integer function.

11. Evaluate $\lim\limits_{x\to y}\dfrac{\sin^2 x-\sin^2 y}{x^2-y^2}$

12. Evaluate $\lim\limits_{h\to0}\dfrac{(a+h)^2\sin(a+h)-a^2\sin a}{h}$

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Answer Key

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	D	A	A	D	2	0	2.00	-2

11. $\left(\dfrac{\sin 2y}{2y}\right)$ **12.** $a^2\cos a+2a\sin a$

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RACE
Regular Analysis Through Continuous Exercise

Limits: RACE-03

Single Correct Type Question

1. If $f(x)=\left(\frac{x^2+5x+3}{x^2+x+2}\right)^x$ then value of $\lim_{x\to\infty} f(x)$ is

(A) e^2 (B) e^4 (C) e^6 (D) e

2. $\lim_{x\to 0}\left(\frac{\sin x}{x}\right)^{\frac{\frac{\sin x}{x}}{1-\frac{\sin x}{x}}}$ is equal to

(A) e^{-1} (B) 1 (C) 0 (D) e^2

3. $\lim_{x\to 0}(1+x)^{\operatorname{cosec} x}$ is equal to

(A) e (B) e^2 (C) e^{-1} (D) 1

4. If x_1 and x_2 are the real and distinct roots of $ax^2+bx+c=0$, then value of

$\lim_{x\to x_1}\left(1+\sin\left(ax^2+bx+c\right)\right)^{\frac{1}{x-x_1}}$ is

(A) $e^{a(x_1+x_2)}$ (B) $e^{(x_1-x_2)}$ (C) e^a (D) $e^{a(x_1-x_2)}$

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5. $\lim_{x\to 0}\left(\frac{a^x+b^x+c^x}{3}\right)^{2/x}$; $(a,b,c>0)$ is equal to

(A) $(abc)^{\frac{1}{3}}$ (B) $\frac{1}{3}$ (C) $(abc)^{\frac{2}{3}}$ (D) $\frac{(abc)^{\frac{2}{3}}}{3}$

6. $\lim_{x\to 0}(1-2x)^{1/x}$ is equal to

(A) e^{-2} (B) e (C) e^2 (D) e^{-1}

7. $\lim_{x\to 1} x^{\cot \pi x}$ is equal to

(A) $e^{\frac{2}{\pi}}$ (B) e (C) 1 (D) $e^{\frac{1}{\pi}}$

Multiple Correct Questions

8. Which of the following is correct

(A) $\lim_{x\to\infty} x\{\ell n(x+a)-\ell nx\}=e^a$ (B) $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{pn+q}=e^{p-q}$

(C) $\lim_{x\to 0}\left(1+\tan^2\sqrt{x}\right)^{\frac{1}{2x}}=e^{\frac{1}{2}}$ (D) $\lim_{x\to\infty}\left(\frac{x+6}{x+1}\right)^{x+4}=e^5$

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Integer Type Questions

9. $\lim\limits_{x\to\infty}\left(\dfrac{1+5x^2}{1+3x^2}\right)^{-x^2}$ is equal to

10. $\lim\limits_{n\to\infty}\left(4^n+5^n\right)^{1/n}$ is equal to

11. $\lim\limits_{x\to\infty}\left(\dfrac{x^2+x-1}{3x^2+2x+4}\right)^{\frac{3x^2+x}{x-2}}$ is equal to

12. $\lim\limits_{x\to 0}(\cos x)^{\cot x}$ is equal to

13. If $\lim\limits_{x\to\infty}\left[\dfrac{2x^2+3}{2x^2+5}\right]^{8x^2+3} = e^{-k}$, then the value of k is

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Answer Key

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	A	D	C	A	D	C,D	0	5
Question	11	12	13							
Answer	0	1	8							

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RACE
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Limits: RACE-04

Single Correct Type Question

1. $\lim\limits_{x\to 0}\dfrac{e^x-1-x}{x^2}$ is equal to

(A) $\dfrac{1}{2}$ (B) 2

(C) e (D) 4

2. $\lim\limits_{x\to 0}\dfrac{\sin x-x+\dfrac{x^3}{6}}{x^5}$ is equal to

(A) $\dfrac{1}{5}$ (B) $\dfrac{1}{30}$

(C) $\dfrac{1}{120}$ (D) 0

3. The value of $\lim\limits_{h\to 0}\dfrac{\log_e(1+2h)-2\log_e(1+h)}{h^2}$ is

(A) 0 (B) –1

(C) 1 (D) does not exist

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4. The value of $\lim\limits_{x\to 0}\dfrac{e^x\sin x-x-x^2}{x^3}$ is

(A) 1 (B) 0

(C) 1/3 (D) 2/3

5. $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x\log(1+x)}$ is equal to

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{3}$

(C) 0 (D) 1

6. $\lim\limits_{x\to 0}\left(\dfrac{1}{x^2}-\dfrac{1}{\sin^2 x}\right)$ is equal to

(A) 1/3 (B) 1/6

(C) –1/3 (D) –1/6

7. $\lim\limits_{x\to 0}\dfrac{(1+x)^n-1}{x}$ =_____

(A) n (B) e

(C) 0 (D) 1

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Matrix Match Type Questions

8. **Column-I** **Column-II**

(P) $\lim\limits_{x\to 0}\dfrac{x-\sin x}{\sin(x^3)}$ (1) $-\dfrac{1}{2}$

(Q) $\lim\limits_{x\to 0}\dfrac{x-\tan^{-1}x}{x^3}$ (2) $-\dfrac{1}{6}$

(R) $\lim\limits_{x\to 0}\dfrac{\sin x-x}{x^3}$ (3) $\dfrac{1}{6}$

(S) $\lim\limits_{x\to 0}\dfrac{1+\sin x-\cos x+\log(1-x)}{x^3}$ (4) $\dfrac{1}{3}$

(A) P→3; Q → 4; R → 2; S→1

(B) P→2; Q → 4; R → 1; S→2

(C) P→3; Q → 2; R → 2; S→4

(D) P→2; Q → 3; R → 4; S→1

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Answer Key

Question	1	2	3	4	5	6	7	8
Answer	A	C	B	C	A	C	A	A

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